

# venu madhav mahadevu

## Senior SDET | Test Automation Architect — Playwright · TypeScript · AI-Augmented Testing

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### PROFESSIONAL SUMMARY

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Test Automation Architect and Senior SDET with 16+ years of experience building enterprise-scale quality platforms, including 13 years on Dell.com e-commerce. Architected a Playwright + TypeScript global test framework spanning 15+ international markets that cut development time by 40%. Certified ScrumMaster with deep CI/CD expertise (GitLab CI, GitHub Actions, Azure DevOps) and a track record of building internal tools adopted by 1,000+ engineers. Maintains a public open-source Playwright/TypeScript framework with live CI reporting, and is expanding into agentic AI testing — building AI agents with Python and Model Context Protocol (MCP) that generate, execute, and triage tests autonomously.

### CORE SKILLS

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**Test Automation:** Playwright (expert), Selenium WebDriver, Page Object Model, spec-driven & data-driven frameworks

**API Testing:** Playwright API testing, REST API validation, Postman, Web API automation

**Languages:** TypeScript, JavaScript, Python, C#, SQL

**AI & Agentic Tooling:** Model Context Protocol (MCP), AI agent development (Python), AI-assisted code generation & defect analysis (Windsurf/Cascade, Claude, GitHub Copilot)

**CI/CD & DevOps:** GitLab CI, GitHub Actions, Azure DevOps, Git, Docker, parallel & cross-browser execution (Chromium, Firefox, WebKit), trace/video capture, device emulation

**Test Reporting & Quality:** Playwright HTML/trace reporting, Allure, JIRA-integrated defect triage; working knowledge of BDD (Cucumber) and performance testing (JMeter, k6)

**Data & Platforms:** MongoDB, ASP.NET, SQL, JIRA, test data management; cloud fundamentals (AWS, Azure)

**Domain Expertise:** Global e-commerce compliance — VAT, GST/PAN (India), CPF/CNPJ (Brazil), trade regulations, multi-ship & multi-currency checkout

### PROFESSIONAL EXPERIENCE

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#### Software Senior Engineer — Dell Technologies *Fremont, CA | Sep 2021 – Present*

- Cut development time 40% across 15+ countries by architecting a Playwright + TypeScript global test automation framework for Dell.com, with AI-assisted code generation integrated into the workflow.
- Eliminated per-country test duplication with a spec-driven, data-driven architecture generating 100+ test variants (Cart, Shipping, Payment, Verify & Submit, Thank You) from configuration arrays.
- Validated complex compliance flows — VAT/special invoices, GST/PAN (IN), CPF/CNPJ (BR), trade compliance, and multi-ship orders — across US, JP, KR, CN, MY, UK, IT, and more.
- Integrated MongoDB for test-result persistence and built AI-augmented defect analysis tied to JIRA, streamlining triage and bug reporting at scale.
- Configured GitLab CI pipelines for automated cross-browser execution; mentor engineers on Playwright, Selenium, and AI-augmented development practices.

#### Associate Software Architect — UST Global (Dell contract) *Fremont, CA | Mar 2020 – Sep 2021*

- Led automation architecture for Dell's e-commerce platform, driving the transition to scalable test infrastructure and CI/CD continuous-delivery pipelines.
- Built Test Data Management tooling adopted across multiple Dell teams, including the Premier team rollout.
- Automated Order Status application testing, supporting multiple product launches with zero critical defects.

#### Career Break — Family responsibilities *| Dec 2018 – Mar 2020*

- Took planned time away to support family; relocated and returned to full-time work at Dell (via UST Global) in March 2020.

#### Software Senior Engineer — Dell Technologies *Bangalore, India | Jun 2015 – Dec 2018*

- Built the ROM application test framework with parallel service calls; earned the Team Award and CSAT Award.

- Created the Host Entry Change tool adopted by 1,000+ engineers across eSupport, Premier, UAT, and E2E teams; presented at the Cloud IT Program.
- Won Q1 Silver Award for outstanding performance; mentored the team on Web API automation and Selenium WebDriver.

**Software Programmer** — Capgemini India (Dell contract)*Bangalore, India | Nov 2012 – Jun 2015*

- Developed and maintained automated test suites for Dell e-commerce applications using Selenium WebDriver and C#.

**Software Programmer** — C4U IT Solutions*Bangalore, India | Nov 2010 – Nov 2012*

- Built web applications and automated testing solutions for enterprise clients using ASP.NET and C#.

**Software Engineer** — Eduquity Career Technologies*Bangalore, India | Feb 2010 – Oct 2010*

- Developed features and tests for education-technology web applications.

## OPEN-SOURCE PROJECTS

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**Automation Exercise E2E Test Framework** — Playwright · TypeScript · GitHub Actions · Allure [GitHub](#) | [Live Allure report](#)

- Built a production-style E2E framework with Page Object Model and custom Playwright fixtures — dependency-injected page objects, an ApiClient, and account setup/teardown lifecycle fixtures.
- Implemented genuine hybrid API + UI testing: full REST endpoint coverage including negative/error contracts, with cross-layer assertions (API creates state, UI verifies, and vice versa).
- Runs a cross-browser CI matrix (Chromium, Firefox, WebKit) on GitHub Actions, publishing a live Allure report to GitHub Pages with screenshots, video, and traces on failure.
- Engineered reliability at the network layer — blocking third-party ad domains that broke waits and intercepted clicks, instead of masking flakiness with retries.

**Agentic AI Test Agent (in progress)** — Python · Model Context Protocol (MCP) · Playwright

- Building an MCP server in Python that lets AI agents generate, execute, and triage Playwright tests autonomously — working toward self-healing test frameworks.

## CERTIFICATIONS & EDUCATION

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Certified ScrumMaster® (CSM) — Scrum Alliance (Cert ID: 000791026)

Master of Computer Applications (MCA) — Andhra University, Visakhapatnam

Bachelor of Science (B.Sc.) — Andhra University, Visakhapatnam